

Programme: B E – Information Science & Engineering
Internal Assessment – II

TERM : 3 rd March to 21 st June 2025	COURSE NAME: Microcontrollers
DATE : 16-06-2025 TIME: 9.15 to 10.15am	COURSE CODE : IS42
MAX MARKS: 30	PORTIONS : L20- L42



Mobile Phones are banned

Instructions to Candidates: **Answer any TWO full questions.**

Marks: 15x2=30

Q. NO	Questions	Blooms Levels (L1 to L6)*	CO	Marks
1.a)	Write the block diagram of a typical ARM7 Processor.	L1	CO3	03
b)	Write a C program to display the string 'ISE Dept' in the serial window of UART1.	L3	CO4	06
c)	Discuss the special registers of cortex-M0.	L2	CO5	06
2.a)	Find the output of the following instructions given that R2=0x0F, R1=0xF0. AND R3, R1, R2 ANDS R3, R1, R2 BIC R5, R1, R2	L3	CO3	03
b)	Explain the important features of the LPC214x series.	L2	CO4	06
c)	Explain Modes and States of Cortex M0 with suitable diagram .	L2	CO5	06
3.a)	Explain any 2 ARM7 Assembler directives.	L2	CO3	03
b)	Write a C program to generate an asymmetric square wave of 120Hz and have a duty cycle of 25% using the Timer0 module.	L3	CO4	06
c)	Discuss the various interrupts supported by the Cortex-M0 processor.	L2	CO5	06

* L1 – Remember, L2 – Understand, L3- Apply, L4- Analyze, L5-Evaluate, L6-Create